# Exploring Patterns of Environmental Justice

**EDS 223 Homework 2**

Roshan Andres Nair

```
# Library of Packages
library(tidyverse)
library(sf)
library(here)
library(stars)
library(tmap)
library(quarto)
library(gt)
library(cowplot)
```

```
# Set Theme
theme_set(theme_bw())
```

## Part 1: Legacy of redlining in current environmental (in)justice

Exploration of historical redlining in Los Angeles and its legacy on present-day environmental justice.

```
# Read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(
  "data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb",
  quiet = TRUE
)

# Filter to Los Angeles County, CA
la_co <- ejscreen |>
  dplyr::filter(CNTY_NAME %in% c("Los Angeles County"))
```

```r
# Read in mapping inequality data for Los Angeles
mi_la <- sf::st_read(
  "data/mapping-inequality/mapping-inequality-los-angeles.json",
  quiet = TRUE
)

# Fixing HOLC geometries
holc <- st_make_valid(mi_la)
```

```r
# Transfrorming mapping inequality CRS
mi_la_transform <- st_transform(mi_la, crs = st_crs(la_co))

# Checking for matching CRS
if (st_crs(la_co) == st_crs(mi_la_transform)) {
  print('Match')
} else {
  warning('NOT A MATCH')
}
```

```
[1] "Match"
```

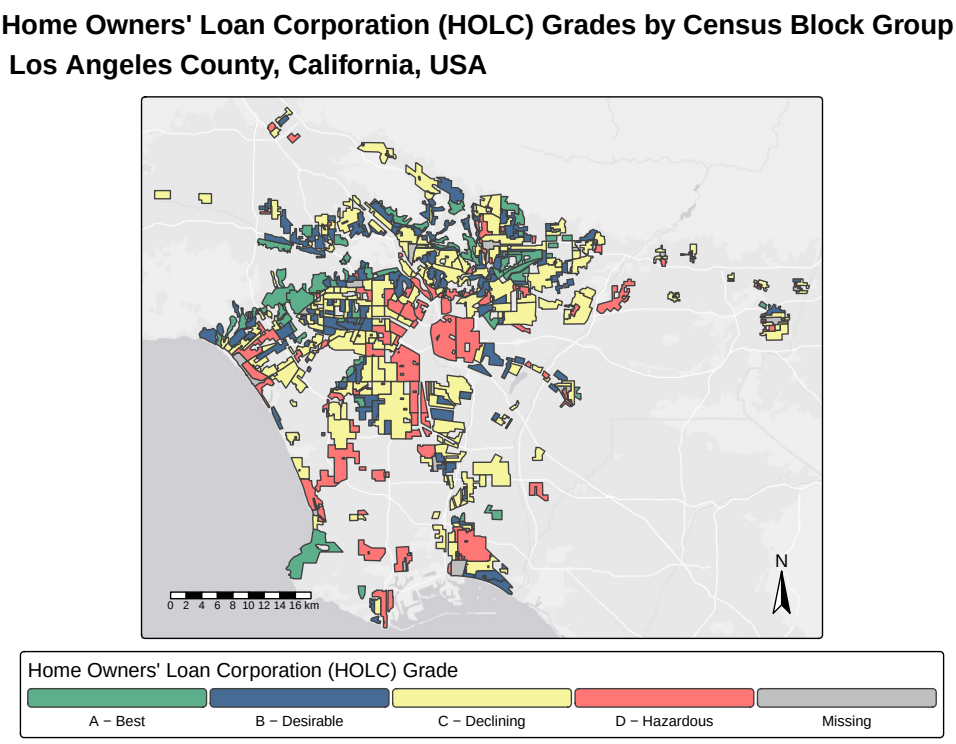
```r
# Joining mapping inequality and LA EJScreen
joined_la <- st_join(la_co, mi_la_transform)
```

1. Mapping of historical redlining neighborhoods:

```r
# Setting colors for HOLC grade levels and text for HOLC grade labels
grade_colours <- c('#5DAE8B', '#466C95', '#F6F49D', '#FF7676', '#9b9090')
grade_labels <- c('A - Best', 'B - Desirable', 'C - Declining', 'D -
 ↪  Hazardous')
```

```r
# Mapping historically redlined neighbourhoods (HOLC Grade by Census Block
 ↪  Group)
tm_shape(mi_la) + # HOLC polygons
  tm_polygons(
    fill = 'grade',
    fill.scale = tm_scale_categorical(
      labels = grade_labels,
      values = grade_colours
    ),
```

```
  fill.legend = tm_legend(
    title = "Home Owners' Loan Corporation (HOLC) Grade",
    orientation = 'landscape',
    width = 50
  )
) +
tm_scalebar(position = c('left', 'bottom')) +
tm_compass() +
tm_title(
  "Home Owners' Loan Corporation (HOLC) Grades by Census Block Group \n Los
   ↪  Angeles County, California, USA",
  size = 1.25,
  fontface = 'bold'
) +
tm_layout(component.autoscale = FALSE) +
tm_shape(la_co) + # LA County basemap
tm_basemap('Esri.WorldGrayCanvas')
```




2. Table summarizing percentage of census block groups that fall within each HOLC grade:

Percent Census Blocks with each HOLC Grade

| Grade | Percent |
|---|---|
| A | 5.00 |
| B | 13.79 |
| C | 34.02 |
| D | 14.98 |
| NA | 32.22 |

```
# Creating dataframe for percent of block groups in each HOLC grade
percent_block_grade <- joined_la |>
  st_drop_geometry() |>
  count(grade) |>
  mutate(pct_grade = round(n / sum(n) * 100, 2)) |>
  select(-n) |>
  rename(Grade = grade, Percent = pct_grade)

# Displaying dataframe as a table
percent_block_grade |>
  gt() |>
  tab_header(
    title = 'Percent Census Blocks with each HOLC Grade'
  )
```

3. Visualization summarizing current conditions within HOLC grades using the mean of the following variables % low income, percentile for Particulate Matter 2.5, and percentile for low life expectancy:

```
# Creating summary dataframe containing means of each current condition
 ↪  variable by grade
summary_la <- joined_la |>
  group_by(grade) |>
  drop_na(LOWINCPCT, P_PM25, P_LIFEEXPPCT) |>
  summarise(
    avg_lowinc = mean(LOWINCPCT),
    avg_pm25 = mean(P_PM25),
    avg_lifexpv = mean(P_LIFEEXPPCT)
  )
```

```r
# Plotting summary of means of each current condition variable by grade

# Create fake plot for legend
mock_plot <- summary_la |>
  ggplot(aes(x = grade, y = avg_lifexpv, fill = grade)) +
  geom_col() +
  scale_color_manual(
    name = 'HOLC Grade',
    values = grade_colours,
    labels = grade_labels
  ) +
  scale_fill_manual(
    name = 'HOLC Grade',
    values = grade_colours,
    labels = grade_labels
  ) +
  theme(
    legend.title = element_text(size = 12, face = 'bold'),
    legend.text = element_text(size = 10),
    legend.key.size = unit(0.5, 'cm')
  )

# Create joint legend using fake plot
legend <- get_legend(mock_plot)

# Low income plot
inc <- summary_la |>
  ggplot(aes(x = grade, y = avg_lowinc, fill = grade)) +
  geom_col() +
  labs(
    title = 'Mean Low Income % by Grade',
    x = 'HOLC Grade',
    y = 'Mean Low Income %'
  ) +
  scale_fill_manual(values = grade_colours) +
  theme(legend.position = 'none')

# PM 2.5 plot
pm <- summary_la |>
  ggplot(aes(x = grade, y = avg_pm25, fill = grade)) +
  geom_col() +
  labs(
    title = 'Mean PM 2.5 Percentile by Grade',
```

```
    x = 'HOLC Grade',
    y = 'Mean PM 2.5 Percentile'
  ) +
  scale_fill_manual(values = grade_colours) +
  theme(legend.position = 'none')

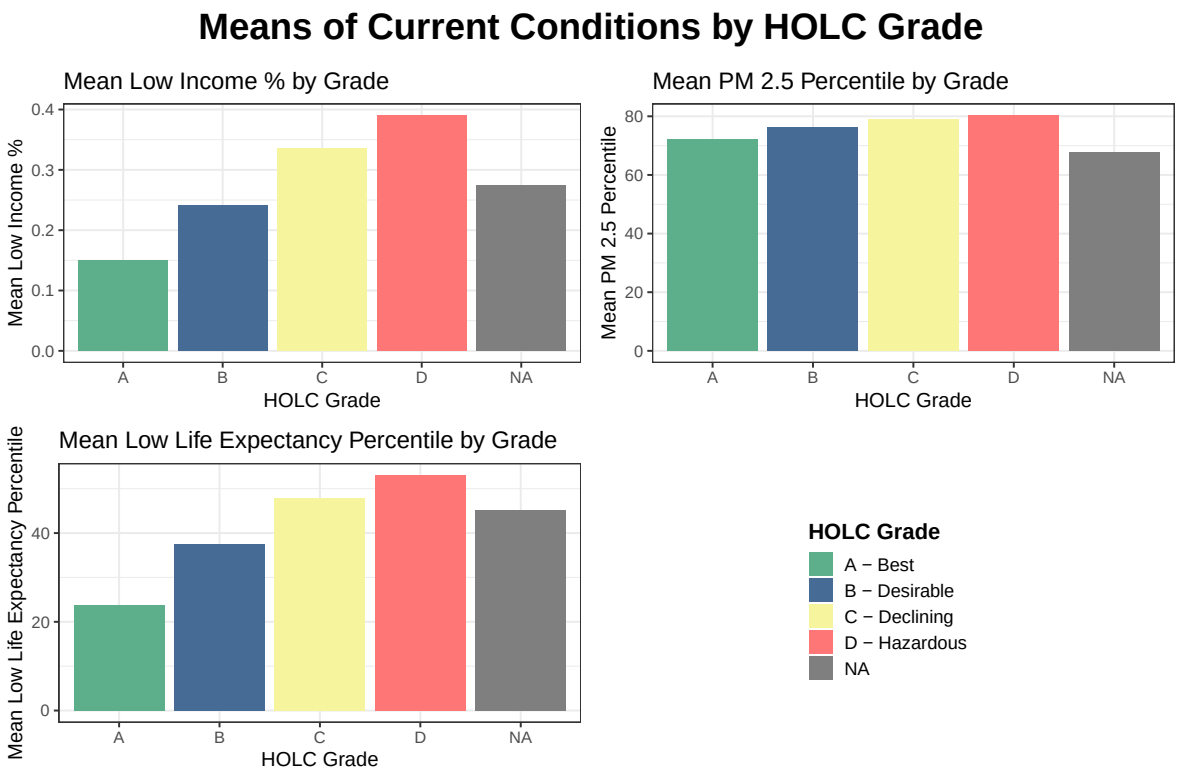
# Left expectancy plot
lex <- summary_la |>
  ggplot(aes(x = grade, y = avg_lifexpv, fill = grade)) +
  geom_col() +
  labs(
    title = 'Mean Low Life Expectancy Percentile by Grade',
    x = 'HOLC Grade',
    y = 'Mean Low Life Expectancy Percentile'
  ) +
  scale_fill_manual(values = grade_colours) +
  theme(legend.position = 'none')

# Joining 3 plots and legend
current_conditions <- plot_grid(
  inc,
  pm,
  lex,
  legend
)

# Plot title
title <- ggdraw() +
  draw_label(
    'Means of Current Conditions by HOLC Grade',
    fontface = 'bold',
    size = 20,
    hjust = 0.5
  )

# Adding title to current conditions plot, changing relative sizes
current_conditions_summary <- plot_grid(
  title,
  current_conditions,
  ncol = 1,
  rel_heights = c(0.1, 1)
)
```

```
# Display summary plot
current_conditions_summary
```




4. Reflecting on the patterns observed in results:

**With each variable describing current conditions, there is a rise in mean values from HOLC grades A to D, where grade D has the highest mean low income percentage, PM 2.5 percentile, and low life expectance percentile, and grade A has the lowest. The means for low income percentage and low life expectance percentile are larger for census blocks in LA County without a grade than those in grades A and B, however the mean for PM 2.5 percentile is lowest in census blocks with no grade designation. The mean PM 2.5 percentile is likely the lowest in ungraded census blocks, as the majority of graded blocks are located within the relatively denser urban sprawl of Los Angeles, whereas there are large numbers of ungraded blocks in less-populated, less dense areas of the county. As a census block is graded more poorly and redlined, the higher the census blocks current condition marker variables. Therefore, historically redlined neighbourhoods are experiencing the effects in the present with reduced life expectancy, lower incomes, and higher PM 2.5.**

## Part 2: Legacy of redlining in pedstrian and bike accidents

Exploration of the legacy of historical redlining in Los Angeles on the report of pedestrian and bike crashes/ fatalities.

1. Summary of the percent of observations within redlined neighborhoods within each HOLC grade:

```
# Read in data, convert to tibble, drop NA values from coodinate cols, rename
 ↪  coordinate cols, convert to spatial features
crashes <- read_csv(here::here("data", "SWITRS_LA", "Crashes.csv")) |>
  as_tibble() |>
  drop_na(POINT_X, POINT_Y) |>
  rename(long = POINT_X, lat = POINT_Y) |>
  st_as_sf(coords = c("long", "lat"), crs = 4326, remove = FALSE)
```

```
# Check that crs match before joining
if (st_crs(holc) == st_crs(crashes)) {
 print('Match')
} else {
 warning('NOT A MATCH')
}
```

[1] "Match"

```
# Join crashes and holc, drop geometry, filter for bicycle and pedestrian
 ↪  accidents
crashes_holc <- st_join(holc, crashes, join = st_intersects) |>
 ↪  st_drop_geometry() |> filter(BICYCLE_ACCIDENT == 'Y'| PEDESTRIAN_ACCIDENT
 ↪  == 'Y')
```

```
# Create summary dataframe of accident counts by grade and respective
 ↪  percentages
crashes_holc_summary <- crashes_holc |>
  group_by(grade) |>
  summarise(count = n()) |>
  mutate(percent = count / sum(count) * 100)

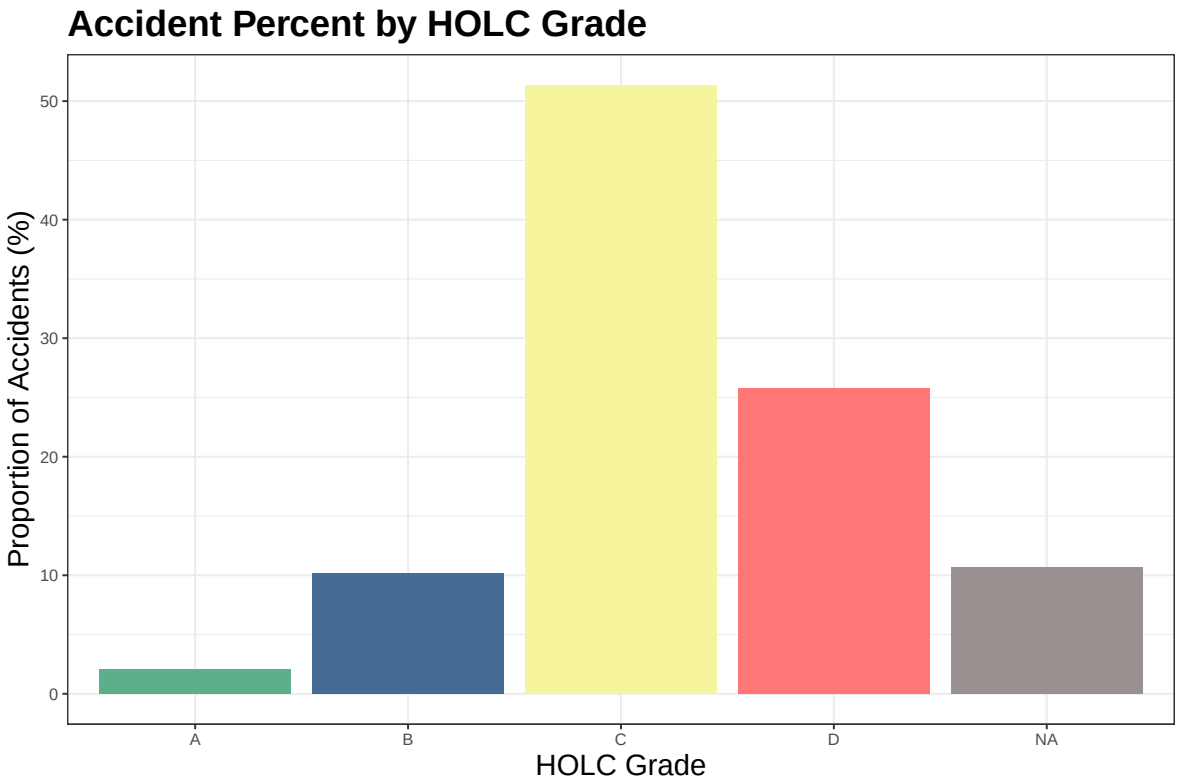
# Display summary df as table
gt(crashes_holc_summary) |>
  tab_header(
```

Accident Counts by HOLC Grade and Percentages

| grade | count | percent |
|---|---|---|
| A | 360 | 2.048947 |
| B | 1782 | 10.142288 |
| C | 9026 | 51.371656 |
| D | 4532 | 25.793967 |
| NA | 1870 | 10.643142 |

```
    title = 'Accident Counts by HOLC Grade and Percentages'
  )
```

```
# Plotting crash summary
ggplot(crashes_holc_summary, aes(x = grade, y = percent)) +
  geom_col(fill = grade_colours) +
  labs(
    title = 'Accident Percent by HOLC Grade',
    x = 'HOLC Grade',
    y = 'Proportion of Accidents (%)'
  ) +
  theme(
    plot.title = element_text(face = 'bold', size = 20),
    axis.title = element_text(size = 16)
  )
```


Accident Percent by HOLC Grade
Proportion of Accidents (%)
50
40
30
20
10
0
A
B
C
D
NA
HOLC Grade